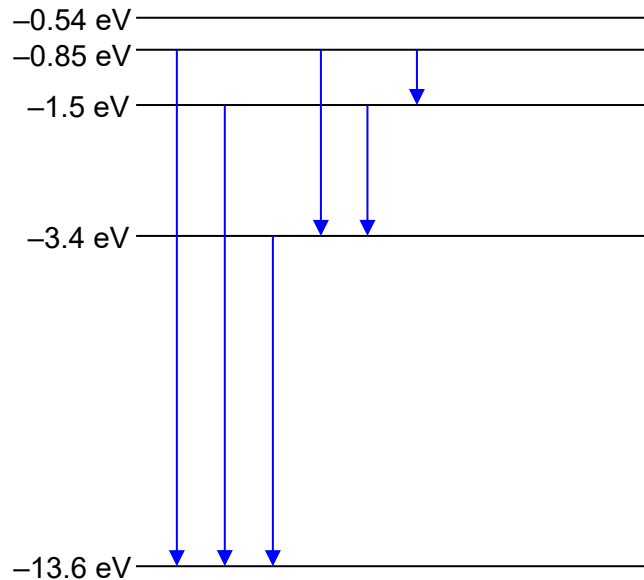


- 9 (a) (i) Particulate: photoelectric effect [B1]
Wave: diffraction / interference / superposition [B1]

- (ii) When electrons transit from higher to lower energy levels, they emit a photon whose energy is equal to the energy difference between the two levels. [B1]

Since the energy levels have discrete (specific) energies, the photon emitted will have specific energies, and hence specific frequencies. Giving rise to a line emission spectrum. [B1]

- (b) (i)



Gas atoms can be excited up to the -0.85 eV level. Hence a total of 6 transitions are possible.

[B1] show understanding that highest possible excited level is -0.85 eV

[B1] draw all six transitions correctly, arrows pointing down

- (ii)

$$h \frac{c}{\lambda} = \Delta E$$

$$\left(6.63 \times 10^{-34}\right) \frac{c}{\lambda} = (13.6 - 0.85) \left(1.60 \times 10^{-19}\right) \quad [\text{M1} - \text{correct energy levels, M1} - \text{correct sub}]$$

$$\lambda = 9.75 \times 10^{-8} = 9.8 \times 10^{-8} \text{ m} \quad [\text{A0}]$$

- (c) (i) Photoelectrons are emitted from metal Z with a range of kinetic energies. [B1]

As the collector plate is made more and more negative, more and more photoelectrons will not have sufficient energy to overcome the potential difference, [B1]

OR electrons experience a larger and larger repulsive force from the collector plate,

and hence less and less electrons will reach the collector plate, causing a gradual decrease in the current.

- (ii) Maximum current = 2.7 mA

$$\frac{N}{t} = \frac{I}{e} = \frac{2.7 \times 10^{-3}}{1.6 \times 10^{-19}} \quad [\text{M1}]$$

$$= 1.69 \times 10^{16} \text{ s}^{-1} \quad [\text{A1}]$$

- (iii) $E_{k,\text{max}} = eV_s = (1.6 \times 10^{-19})(5.3) = 8.48 \times 10^{-19} \text{ J} \quad [\text{M1}]$

$$h \frac{c}{\lambda} = \Phi + E_{k,\text{max}}$$

$$\left(6.63 \times 10^{-34}\right) \frac{3.00 \times 10^8}{9.75 \times 10^{-8}} = \Phi + 8.48 \times 10^{-19} \quad [\text{M1}]$$

$$\Phi = 1.19 \times 10^{-18} \text{ J} \text{ or } 1.18 \times 10^{-18} \text{ J (if use } 9.8 \times 10^{-8})$$

$$= 1.2 \times 10^{-18} \text{ J} \quad [\text{A0}]$$

- (iv).

$$\Phi = h \frac{c}{\lambda_0}$$

$$1.2 \times 10^{-18} = \left(6.63 \times 10^{-34}\right) \frac{3.00 \times 10^8}{\lambda_0}$$

$$\lambda_0 = 1.66 \times 10^{-7} \text{ m}$$

$$= 166 \text{ nm} \quad [\text{B1}]$$

$$\text{or } 167 \text{ nm (if use } 1.19 \times 10^{-18})$$

$$\text{or } 169 \text{ nm (if use } 1.18 \times 10^{-18})$$

- (v) Since the stopping potential is now larger, the energy of the photons incident on the metal Z is larger. [M1]

Hence the EM radiation from the new source has a higher frequency. [A1]

- (vi) Photocurrent increases with intensity of light. [B1 for stating the characteristic]

Hence light meters can use the amount of photocurrent to detect the ambient light intensity. [B1 for how the characteristic is used]

- (vii) The threshold wavelength is smaller than that of visible light, hence visible light will not cause emission of photoelectrons. [B1]